Using Spreadsheets

Purpose

The purpose of this pre-lab exercise is to introduce the basic spreadsheet skills required in this course and to provide some practice with those skills. We will be using Google Sheets in this course, so that groups can collaborate during labs. You should use Google Sheets to complete this lab. If you are familiar with Excel (or another spreadsheet program), much of this will be familiar, though the exact commands and menu options will be different.

This exercise is intended to be done individually. You should complete it before your first lab.

Training Videos

Before beginning this practice, please watch the three training videos linked below.

Video 1 Video 2 Video 3

If you are familiar with the basics of entering and formatting text and numbers in a spreadsheet, you can skip to Video 2. If you are familiar with the basics of entering functions in spreadsheets (finding sums, averages, medians and so on of a list of numbers), you can skip to Video 3. Everyone should watch Video 3 unless you are very proficient with spreadsheets and have used them extensively in the past to do advanced work.

Allowed Functions and Operations

Throughout this exercise, you should use arithmetical operators ($+$, $-$, $*$, $/$). You may ***not*** use functions like MINUS, MULTIPLY, DIVIDE, etc. You must also not use array functions. In short, you should use only ideas covered in the training videos.

Instructions

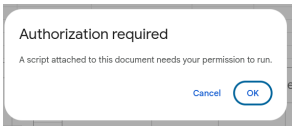
Read all these instructions carefully and follow them all!

To make a copy of the spreadsheet for this lab, click on the link below. You will be prompted to make a copy of the spreadsheet. You should rename the copy with your name and save it in a folder on your Google Drive.

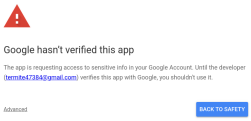
Spreadsheet

This spreadsheet auto-grades itself. When you have done the work (or even just part of it), you can grade the sheet by clicking on the 'Grade My Lab!' button in either tab of the spreadsheet.

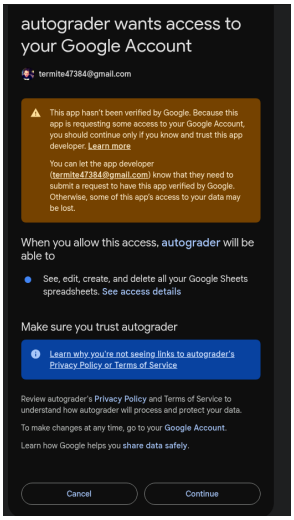
The first time you do this, you will be prompted to authorize the autograder:



Click 'OK'



Click 'Advanced'



Click 'Continue'

When you do this, the sheet will grade itself and give you both a score and detailed feedback on the 'Feedback' tab. You can regrade the sheet as many times as you like. You must get all the points to get credit for this assignment.

Once you have completed the lab, go to the Gradescope assignment for it. The assignment will ask you to enter the URL of your spreadsheet. You can get this by clicking on the 'Share' button in the upper right corner of the spreadsheet, and copying the link that appears. You should make sure that the link you copy allows anyone with the link to view your spreadsheet. Paste this link into the Gradescope assignment and submit it.

As always, by submitting the assignment, you are certifying that you did all the work on your own.

Part I: Functions, Copying, and Scatter Plots

You will see the following data in Columns A, B, and C of the first tab.

t	d	v
1.2	7.056	11.76
3.4	56.64	33.32
4.1	82.369	40.18
4.9	117.649	48.02
5.3	137.64	51.94
7.1	247.009	69.58

1. Make two scatter plots; one for each of the following pairs of lists.

   (a) v vs. t; (i.e., put v on the vertical axis, and t on the horizontal axis)

   (b) v vs. d

2. In Column D, insert and copy down formulas to compute $\frac{v}{t}$. Be sure you are copying formulas rather than typing a new formula in each cell of the column. If you did this correctly, then there will be a noticeable feature of the numbers in this column.

3. In Column E, Make a list whose entries are the differences of consecutive elements of the d (which is in Column B). That is, Cell $E2$ should contain the difference of Cell $B3$ and $B2$. Insert this formula, and copy it down. You should have one less entry in Column E than in Column B. Why? We denote these new values by Δd.

4. Do the same for t (contained in Column A) in Column F to compute Δt.

5. Next, compute $\frac{\Delta d}{\Delta t}$ in Column G. Your column should contain five entries, the first of which is about 22.5382.

Part II: Scatter Plots, Linear Trends, and Line Plotting

In the second tab of the spreadsheet for this lab, you will find data for some x and y values.

6. First, let's make this data a little easier to read: format column B so that numbers are displayed to two decimal places. (Hint: Use the Format menu.)

7. Create a scatter plot of y vs. x.

8. We will now add a linear trendline to your graph:

   - Open the Chart editor if it isn't already open by double clicking on your chart.
   - Click on 'Customize', then 'Series'.
   - Check the 'Trendline' box, then select 'Use Equation' in the 'Label' pulldown menu.

This data was created by adding some random noise to the line $y = 10x + 7$. As you should be able to see, the trendline is not quite that line. Next, we will generate data for that line, add it to the chart, and see how to play with the slope and y-intercept. These are skills you will need in the first two labs of the semester.

9. In the cells next to the 'Slope' and 'y-intercept' labels ($C13$ and $C14$ respectively), enter the slope and y-intercept of the above line.

10. In Column C, enter a formula to calculate the y values on this line (given the x values in Column A). **Be sure to use the values in the cells you just entered! You should not put '$= 10 * A2 + 7$' in Cell $C2$!**

    Also, when you're copying the formula down to fill Column C, be sure you're still multiplying by the correct slope and y-intercept. If not, or if you are not sure how to use dollar (\$) signs in spreadsheets, you probably want to go back and rewatch the third video.

11. Once again, open the Chart editor by double clicking on your chart. On the 'Setup' tab, click on the little grid next to the 'Add Series'. Select the data in Column C. The points in the column should appear on your chart.

12. The easiest way to add a linear line to a plot is to generate data like you just did, then add a linear trendline for it (it will be the exact right line). Under the 'Customize' tab, change the Series to 'y (modelled)', change the Point size to 'None', the Trendline to 'Linear', and the Label to 'Use equation'. You should see the line added to the chart, and its equation displayed.

13. Now change the slope and y-intercept in Cells $C13$ and $C14$ respectively. **If you did all the above correctly, you should see the line changing on your chart! If not, you may not have followed the instructions in Question 10. Go back, read that question, and redo your formulas in column C.**